

Hemostasis, Thrombosis, & Bleeding Disorders

Natasha M. Savage, MD, FCAP • designed to follow alongside the 50-slide coagulation deck

Official Week 7 LOs

Slide-order companion

6 ScholarRx Bricks

Hemophilia B dedication

Current 2026 treatment enrichment

Factoid sheet companion

Overview: Official Week 7 Learning Objectives

LO 1. Describe primary hemostasis in terms of key components and events, including vWF, platelets, and fibrinogen.

LO 2. Understand testing of primary hemostasis: platelet count, MPV, bleeding time/PFA, and platelet aggregation studies.

LO 3. Understand disorders of primary hemostasis and their work-up, including thrombocytopenia, Glanzmann thrombasthenia, Bernard-Soulier syndrome, and von Willebrand disease.

LO 4. Describe secondary hemostasis, including the major coagulation proteins and vitamin K-dependent proteins.

LO 5. Interpret secondary-hemostasis testing: PT, aPTT, mixing studies, factor assays, and inhibitor titers.

LO 6. Understand secondary-hemostasis disorders and work-up, including hemophilia A, hemophilia B, other factor deficiencies, and acquired factor inhibitors.

LO 7. Build the differential diagnosis of microangiopathic hemolytic anemia and know smear findings, clinical setting, work-up, and management.

LO 8. Describe common congenital and acquired thrombophilias, including Factor V Leiden, prothrombin gene mutation, protein C/S deficiency, and related work-up.

SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

The Coagulation Cascade

Bleeding Disorders: Foundations & Frameworks

Tests for Hemostasis

Hemophilia

Thrombotic Microangiopathies

Thrombotic Disorders: Foundations & Frameworks

LO 1-3

Slides 4-16: platelet plug physiology, vWF/fibrinogen, PFA/aggregation testing, ITP, Glanzmann, Bernard-Soulier.

LO 4-6

Slides 18-32: cascade, PT/aPTT/mixing, hemophilia A/B, other factor defects, vWD.

LO 7

Slides 34-40: DIC, TTP, HUS and the laboratory differences that keep them straight.

LO 8

Slides 42-48: natural anticoagulants, Factor V Leiden, prothrombin mutation, protein C/S deficiency.

SOURCE NOTE

The primary source is Savage's 50-slide coagulation lecture. The uploaded companion is titled **Lectures 6 and 7**; because our Week 7 core sequence reserves this material as **Lecture 6**, the companion is treated as supplementary rather than allowed to renumber the hub. It also expands HIT and several board-style explanations that the slide deck only touches briefly.

THE LECTURE'S MASTER FRAMEWORK

Primary hemostasis = platelet plug. Think mucocutaneous bleeding.

Secondary hemostasis = fibrin reinforcement. Think delayed/deep bleeding, hemarthroses, hematomas.

Thrombophilia = the same system tipped too far toward clotting.

Slides 4-16

Primary hemostasis: adhesion -> activation -> aggregation

LO 1-3

BASICS

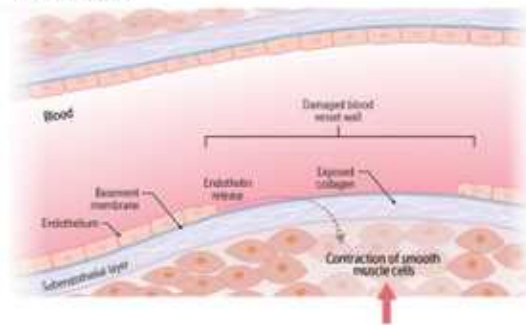
Hemostasis: AKA clotting; how bleeding stops

Includes plasma containing coagulation proteins, vasculature, and platelets

Issues result in hemophilia (bleeding too much) or thrombophilia (clotting too much)

3 steps: **vessel constriction**, platelet plug formation (1° hemostasis), and fibrin formation (2° hemostasis)

A. Vasoconstriction

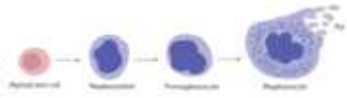


Lecture slide 4 - full slide retained. Savage starts with vessel constriction, then the platelet plug (primary hemostasis), then fibrin stabilization (secondary hemostasis). Keep those layers separate as you move through the deck.

Slides 6-8: key players

- **GpIb** binds **vWF** -> platelet adhesion.
- **P2Y12** binds ADP -> platelet activation/recruitment.
- **GpIIb/IIIa** binds **fibrinogen** -> platelet-to-platelet aggregation.
- Platelet alpha granules contain proteins including fibrinogen and vWF; dense granules contain ADP, serotonin, calcium, etc.
- vWF comes from endothelial Weibel-Palade bodies, platelet alpha granules, and subendothelium; it also carries factor VIII.

PRIMARY HEMOSTASIS: KEY PLAYERS



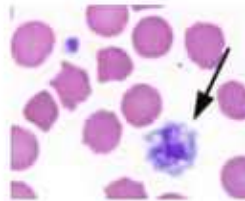
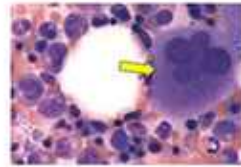
1. Platelets (thrombocytes): fragments of megakaryocyte cytoplasm; thrombopoietin from the liver stimulates megakaryocyte production; "live" for 8-10 days in blood

• Membrane

- Phospholipid surface for coagulation factors
- Glycoprotein Ib (GpIb): binds von Willebrand factor (vWF)
- GpIIb/IIIa: binds fibrinogen
- P2Y₁₂: binds ADP allowing for activation

• Contents

- Alpha granules (purple color on peripheral blood smear): fibrinogen, vWF, others
- Dense granules (electron microscopy needed): serotonin, ADP, calcium, others
- Contractile proteins (located on the outer pale area of the platelet or hyalomere)



Lecture slide 6 - full slide retained. This slide is receptor-heavy and worth memorizing: GpIb-vWF for adhesion, GpIIb/IIIa-fibrinogen for aggregation, and P2Y₁₂-ADP for activation.

PRIMARY HEMOSTASIS: KEY PLAYERS

2. von Willebrand factor (vWf):

In your platelets (alpha granules), endothelial cells (**Weibel-Palade bodies**), subendothelial connective tissue

Huge protein composed of multimers
• Cleaved by ADAMTS13

Adheres platelets to subendothelium
• Via GpIb

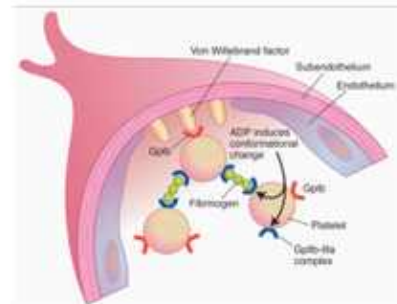
Carries factor VIII (VIII can't circulate alone)

Amount varies

- Low in type O blood group
- Acute phase reactant (increases during inflammatory states)

3. Fibrinogen:

Coagulation protein AKA factor II (more on this later)



Lecture slide 7 - full slide retained. vWF links platelets to exposed subendothelium and carries factor VIII. One slide-text typo calls fibrinogen factor II; later slides correctly identify fibrinogen as factor I.

SLIDE TYPO TO ACTIVELY CORRECT IN YOUR HEAD

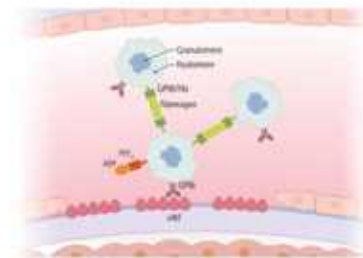
Slide 7 labels fibrinogen as “factor II,” but Slide 18 correctly lists **factor I = fibrinogen/fibrin** and **factor II = prothrombin/thrombin**. Treat the Slide 7 wording as a source typo.

PRIMARY HEMOSTASIS: EVENTS

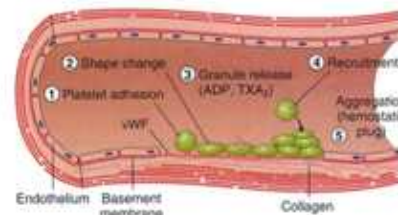
Adhesion: vWf gets exposed and Gp1b binds to it (weak interaction)

Activation: Platelets now release granule contents, recruiting more platelets, and change shape

Aggregation: Finally, GpIIb/IIIa gets expressed and crosslinks with other platelets via fibrinogen



B. PRIMARY HEMOSTASIS



Lecture slide 8 - full slide retained. The three A's are the core primary-hemostasis sequence: adhesion through vWF-GpIb, activation/degranulation, then aggregation through GpIIb/IIIa-fibrinogen bridges.

NEED TO MEMORIZE - THE THREE A'S

Adhesion: vWF - GpIb.

Activation: ADP/P2Y₁₂, TXA₂, granule release, shape change.

Aggregation: GpIIb/IIIa - fibrinogen - GpIIb/IIIa.

Slides 9-11: testing primary hemostasis

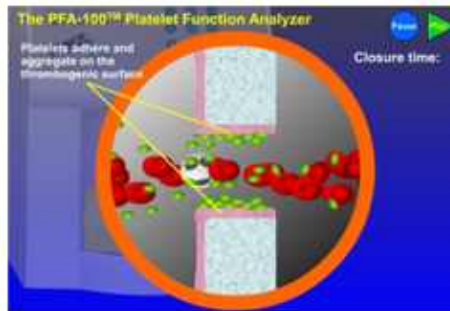
- **Platelet count:** quantitative problem?
- **MPV:** platelet size; larger/younger platelets can suggest peripheral destruction with marrow compensation.
- **Bleeding time:** historical test; Savage notes exams may still ask it.
- **PFA:** more standardized modern platelet-function screening.
- **Aggregation studies:** add agonists such as ristocetin, collagen, ADP, epinephrine and measure the response.

PRIMARY HEMOSTASIS: TESTING

Bleeding time: lancet pierces forearm skin; time it takes from beginning to bleed stopping represents primary hemostasis time; reference range is usually 2-9 minutes

PFA: more standardized bleeding time

- Done at most modern labs including WMCG but tests still ask about bleeding time



Lecture slide 10 - full slide retained. The PFA is presented as a more standardized replacement for bleeding time. Conceptually, it tests how efficiently platelets form a plug under controlled conditions.

Slide 12: what primary-hemostasis bleeding looks like

Petechiae + mucocutaneous bleeding - epistaxis, gingival bleeding, menorrhagia, immediate superficial bleeding - should push you toward platelets/vWF before factor deficiencies.

IMMUNE THROMBOCYTOPENIC PURPURA (ITP): PATHOLOGY

Acquired: presents with isolated thrombocytopenia, often with elevated MPV

Patients produce antibodies against platelet proteins (GpIIb/IIIa) resulting in splenic destruction

Bone marrow will show increased megakaryocytes; megakaryocytes release immature, bigger platelets giving increased MPV

Present with petechiae (1-3 mm) or purpura

Diagnosis of exclusion

Acute:

Children following viral illness
Usually, self-limiting

Chronic:

More common in adults, often women
Primary versus secondary (associated with autoimmune disease, HIV, other infection, malignancy such as CLL, medications)



Lecture slide 14 - full slide retained. ITP presents as isolated thrombocytopenia, often with increased MPV, antibody-mediated splenic destruction, and increased megakaryocytes in marrow. The slide emphasizes petechiae/purpura.

ITP PATTERN

Isolated low platelets + normal PT/aPTT + increased MPV + diagnosis of exclusion.

HEREDITARY PLATELET DISORDERS: PATHOLOGY

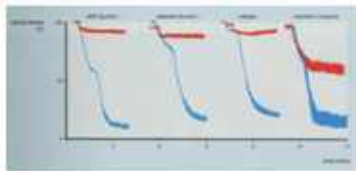
Glanzmann Thrombasthenia

Autosomal recessive

Defect/lack of GpIIb/IIIa

Variable bleeding

Platelet aggregation studies will show lack of aggregation with all agonists but ristocetin



<http://www.practical-haemostasis.com/>

Control: Blue
Patient: Red

Bernard-Soulier Syndrome

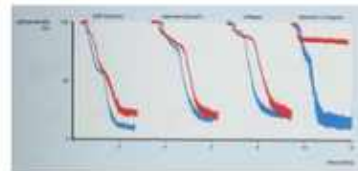
Autosomal recessive

Defect/lack of GpIb

Large platelets with thrombocytopenia

Variable bleeding

Platelet aggregation studies will show lack of aggregation with ristocetin



Lecture slide 16 - full slide retained. Glanzmann = GpIIb/IIIa aggregation defect: abnormal with most agonists but ristocetin preserved. Bernard-Soulier = GpIb adhesion defect: ristocetin response absent, with giant platelets and thrombocytopenia.

Disorder	Defect	Platelets	Ristocetin / aggregation clue
Glanzmann	GpIIb/IIIa	Usually normal count/size	Aggregation fails with ADP/ collagen/epi; ristocetin preserved.
Bernard-Soulier	GpIb	Giant platelets + thrombocytopenia	Ristocetin fails; adding normal plasma does not fix a receptor defect.
vWD	vWF quantity/ function	Usually normal count	Ristocetin-based adhesion abnormal; plasma can supply vWF.

LO 1-3 checkpoint

Can you draw the three A's from memory and distinguish Glanzmann, Bernard-Soulier, ITP, and vWD by platelet count/size and ristocetin behavior?

Slides 18-29

Secondary hemostasis: fibrin, pathways, tests, and factor deficiencies

LO
4-6

SECONDARY HEMOSTASIS: KEY PLAYERS

Coagulation proteins (factors):

Enzymes; most are serine proteases (hydrolyze peptide bonds) with XIII being an exception (crosslinks fibrin); initially inactive in the blood; activate other factors; include II (thrombin), VII, IX, X, XI, XII, and XIII

Co-factors are also important and include III (tissue factor, TF), V, and VIII

Some are vitamin K dependent: II, VII, IX, X (C and S also K dependent*)

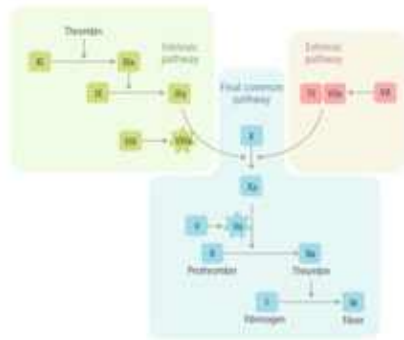
Protein	Important fact
XIII	Transglutaminase-crosslinks fibrin; longest half-life
XII	Not important in vivo; deficiency will not result in bleeding
XI	In vivo, it is the start of intrinsic pathway
X	Vitamin K dependent
IX	Vitamin K dependent
VIII	Co-factor
VII	Vitamin K dependent; shortest half-life
VI	Activated V (Factor Va)
V	Co-factor
IV	Just calcium
III	Co-factor; TF
II	Prothrombin/Thrombin; Vitamin K dependent; second longest half-life
I	Fibrinogen/Fibrin

Lecture slide 18 - full slide retained. Savage's factor table is worth using as a reference: vitamin K-dependent II, VII, IX, X; V and VIII are cofactors; XIII crosslinks fibrin; XII can prolong PTT without causing clinical bleeding.

VITAMIN K DEPENDENT

II, VII, IX, X + proteins C and S.

SECONDARY HEMOSTASIS: EVENTS



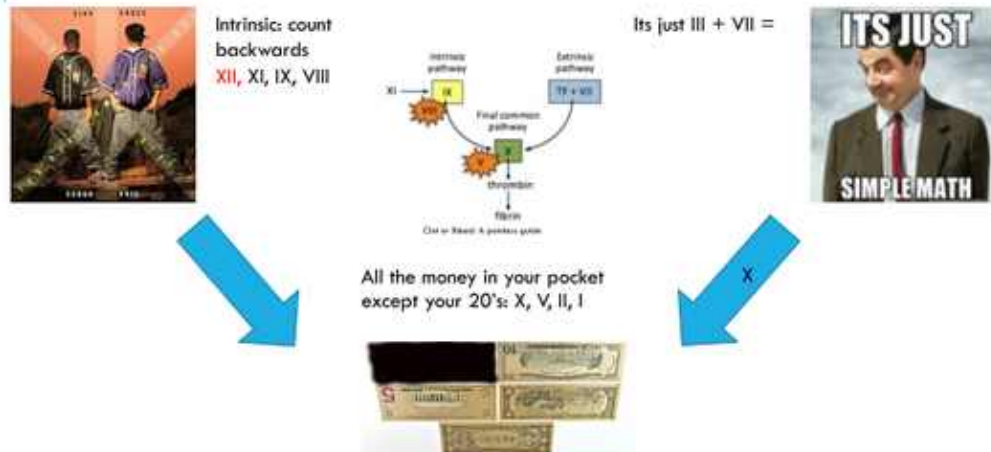
Extrinsic pathway (in vitro something extrinsic must be added: thromboplastin): remember it by simple math or $3(TF)+7=10$

Intrinsic pathway: once a bit of thrombin gets made from above pathway, it activates XI setting this pathway in motion; remember it by counting backwards or 11, 9, 8

Common final pathway ties it all together; remember it by all your money in your pocket less than your \$20s or \$10, \$5, \$2, \$1

Lecture slide 19 - full slide retained. The lecture uses simple numeric mnemonics: extrinsic TF + VII -> X; intrinsic XI -> IX with VIII; common X -> V -> II -> I.

SECONDARY HEMOSTASIS: EVENTS

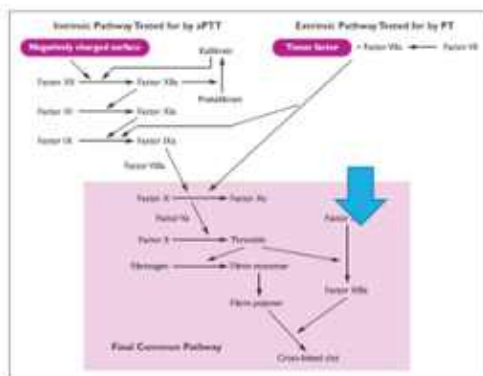


Lecture slide 20 - full slide retained. Savage's slide mnemonic: intrinsic counts backward XII, XI, IX, VIII; extrinsic is III + VII = X; common pathway is X, V, II, I.

PHYSIOLOGY VS TEST-TUBE CASCADE

The classic intrinsic/extrinsic diagram is extremely useful for **interpreting PT and aPTT**, but in vivo coagulation is a network centered heavily around tissue factor and thrombin amplification. Savage explicitly follows the simple cascade with a slide saying it is "really more complicated than that."

XIII: DON'T FORGET HIM



Factor XIII crosslinks fibrin monomers, strengthening clot

It's the Arnold Schwarzenegger of the clotting cascade:

- Long life
- Very strong: crosslinks monomers
- **Elusive***



Lecture slide 21 - full slide retained. Factor XIII acts after fibrin formation to crosslink and strengthen the clot. Because PT/aPTT stop timing once fibrin forms, XIII deficiency can bleed despite normal routine coagulation assays.

Slides 23-25: PT, aPTT, and mixing studies

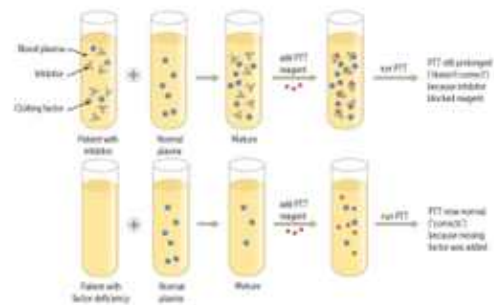
- **PT/INR:** extrinsic + common; especially factor VII sensitivity; used for warfarin monitoring.
- **aPTT:** intrinsic + common; VIII, IX, XI, XII plus common factors; historically used for unfractionated heparin monitoring.
- **Mixing study:** mix patient plasma 1:1 with normal plasma.
- **Correction -> factor deficiency.**
- **No correction -> inhibitor.**
- Factor assay quantifies the suspected factor; inhibitor titer is measured in Bethesda units.

SECONDARY HEMOSTASIS: TESTING

Mixing study: if PT or PTT is prolonged, next step is a mixing study where the patient's plasma is mixed in a 1:1 ratio with normal plasma; if the PT/PTT corrects, a deficiency is diagnosed; if it does not, an inhibitor is diagnosed

Factor assays: measures quantity of factors; so, if PT is prolonged but PTT is normal, you would want to order a factor VII assay to determine extent of deficiency

Inhibitor titers: if you diagnose an inhibitor, you must perform an inhibitor titer to determine strength of the inhibitor; measured in Bethesda units; <5BU is a weak inhibitor



Lecture slide 25 - full slide retained. This is one of the most testable slides. Immediate correction indicates a factor deficiency; persistent prolongation indicates an inhibitor. A time-dependent VIII inhibitor may look different after incubation.

NEED TO MEMORIZE - COAG TESTS

PT = VII + common.

aPTT = XII, XI, IX, VIII + common.

Mix corrects = deficiency. Mix fails = inhibitor.

DISORDERS OF SECONDARY HEMOSTASIS

Patients with disorders of coagulation factors or secondary hemostasis tend to have delayed bleeding after a traumatic event because they have no issues forming a platelet plug; they may bleed into joints or soft tissues (hemarthrosis, hematomas)

Bleeding time/PFA is normal



Lecture slide 26 - full slide retained. Factor deficiencies cause delayed bleeding after trauma and deep bleeding into joints/soft tissues - hemarthroses and hematomas - while primary-hemostasis screening such as PFA may be normal.

Lab pattern	First thought
PT prolonged only	Factor VII deficiency/warfarin-early effect; evaluate extrinsic pathway.
aPTT prolonged only + mixing corrects	VIII, IX, XI, or XII deficiency; clinical bleeding distinguishes XII from hemophilia-type defects.
aPTT prolonged only + mixing fails	Inhibitor such as factor VIII inhibitor or lupus anticoagulant.
PT + aPTT prolonged	Common pathway/multiple-factor issue: DIC, severe liver disease, vitamin K deficiency/warfarin, massive consumption, etc.
Normal PT/aPTT but deep bleeding	Remember factor XIII deficiency can be invisible to routine assays.



A LITTLE DEDICATION

Dedicated to Cole, my real-life Hemophilia B case.

Thanks for making factor IX especially hard to forget. ♥

Slide 27: A and B are clinically very similar

- **Hemophilia A = factor VIII deficiency.**
- **Hemophilia B = factor IX deficiency.**
- Both are classically **X-linked** and produce secondary-hemostasis bleeding: deep tissue bleeding, muscle hematomas, hemarthroses, delayed bleeding after procedures/trauma.
- Typical labs: **normal platelet count, normal PT, prolonged aPTT.**
- If no inhibitor is present, the prolonged aPTT **corrects on mixing**, followed by a low factor VIII or IX assay.

	Hemophilia A	Hemophilia B
Factor	VIII	IX
Pathway	Intrinsic	Intrinsic
PT	Normal	Normal
aPTT	Prolonged	Prolonged
Mixing study	Corrects if simple deficiency	Corrects if simple deficiency
Factor assay	Low VIII	Low IX
Inhibitors in slide	~15%	~2-3%

HEMOPHILIA B COLD SET

Factor IX deficiency -> X-linked -> deep/joint bleeding -> isolated prolonged aPTT -> mixing corrects -> low IX activity.

HEMOPHILIA A AND B: PATHOLOGY



Tube	APTT			
Tube 1 [Normal plasma]	Normal	Normal	Normal	Normal
Tube 2 [Test plasma]	Normal	Prolonged	Prolonged	Prolonged
Tube 3 [Incubated test + normal plasma]	Normal	Normal	Prolonged	Prolonged
Tube 4 [Non-incubated test + normal plasma]	Normal	Normal	Prolonged	Normal
Interpretation	Normal	Clotting factor deficiency	Immediate acting inhibitor	Time-dependent inhibitor *

<http://www.practicalhaematology.com/>

Lecture slide 28 - full slide retained. Savage uses a time-dependent factor VIII inhibitor to show why some inhibitors may require incubation before the mixing study fully fails to correct.

SEVERITY IS FACTOR ACTIVITY, NOT THE LETTER A/B

The slide explicitly ties disease severity to how low factor activity is. A patient with severe B can bleed far more than a patient with mild A; the factor number identifies the protein, while **residual activity predicts phenotype**.

CURRENT 2026 HEMOPHILIA B TREATMENT LANDSCAPE - EXTERNAL ENRICHMENT

The lecture focuses mostly on pathology/work-up, so here is the modern treatment context:

- **Factor IX replacement** remains foundational, including prophylactic recombinant/extended-half-life FIX products.
- **HEMGENIX (etranacogene dezaparvovec)** is an FDA-approved AAV-based gene therapy for selected adults with Hemophilia B who meet the labeled bleeding/prophylaxis criteria.
- **HYMPAVZI (marstacimab)** provides once-weekly non-factor prophylaxis for patients age 12+ with Hemophilia B **without FIX inhibitors**.
- **ALHEMO (concizumab)** is a TFPI antagonist; the current 2025 U.S. label includes age 12+ Hemophilia B **with or without FIX inhibitors**.

FDA HEMGENIX • FDA HYMPAVZI • FDA ALHEMO 2025 label

LO 6 checkpoint - Cole test :)

If someone says **Hemophilia B**, can you immediately say factor IX, isolated prolonged aPTT, normal PT/platelets, correction with mixing if no inhibitor, deep bleeding/hemarthroses, and factor IX assay confirmation?

Slides 31-32

von Willebrand disease bridges primary and secondary hemostasis

LO 1 + LO 3 +
LO 6

VON WILLEBRAND DISEASE: PATHOLOGY

Most common inherited bleeding disorder – 1% of the US

Disease severity is highly variable but may present with increased mucocutaneous bleeding and is therefore diagnosed more in women (due to menses and childbirth)

Mostly affects primary hemostasis as platelets adhere to it, but can also affect secondary hemostasis as factor VIII can't survive without it

Test:

- vWf antigen assay: measure quantity
- Ristocetin activity: measures function
- RIPA: measure response to different quantities of ristocetin
- aPTT/factor VIII assay: measures VIII levels

Lecture slide 31 - full slide retained. vWD is the most common inherited bleeding disorder. It primarily impairs platelet adhesion but can also lower factor VIII, so severe disease may prolong aPTT.

VWD DUAL ROLE

Primary: vWF helps GpIb-mediated adhesion.

Secondary: vWF carries/stabilizes factor VIII.

Testing in the slide

- **vWF antigen** = quantity.
- **Ristocetin activity** = function.
- **RIPA** = platelet-vWF response at different ristocetin concentrations.
- **aPTT / factor VIII assay** can show the secondary-hemostasis effect.
- vWF is an acute-phase reactant and tends to be lower in type O blood, so borderline type 1 disease can require repeat testing.

VON WILLEBRAND DISEASE: PATHOLOGY

Many types due to many mutations

Quantitative: type 1 (autosomal dominant, most common) and type 3 (autosomal recessive, rare; severe; increased PTT due to low factor VIII)

- It is important to know that vWf levels vary greatly based on sex, acute phase, blood type (lowest in type O), and therefore type 1 can be difficult to diagnose; repeat testing may be needed

Qualitative: type 2 to include A, B, M, and N subtypes

- A: loss of high molecular weight multimers; increased cleavage of vWf by ADAMTS13 and retention of vWf in the endothelium
- B: also, loss of high molecular weight multimers; hyperfunctional mutation causing it to bind too tightly to Gplb; a small amount of ristocetin will induce adhesion (RIPA enhanced)

Lecture slide 32 - full slide retained. Type 1 is partial quantitative deficiency and most common; type 3 is severe quantitative deficiency; type 2 subtypes are qualitative and differ by multimer/receptor behavior.

vWD checkpoint

Can you explain why vWD usually looks mucocutaneous but can also raise aPTT, and distinguish a vWF quantity assay from a function assay?

Slides 34-40

MAHA: DIC vs TTP vs HUS

LO 7

Slide 34: all three can share MAHA + thrombocytopenia

The differentiator is **what formed the microthrombi and whether coagulation factors are being consumed.**

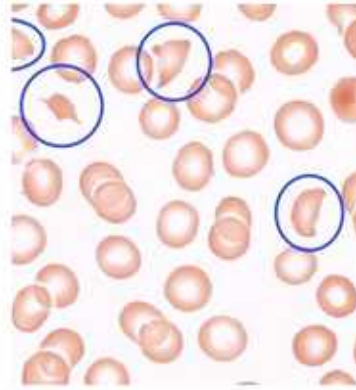
DISSEMINATED INTRAVASCULAR COAGULATION: PATHOLOGY

These small thrombi shear red cells making schistocytes – causing anemia and evidence of hemolysis (increased LDH, low haptoglobin, increased reticulocytes, increased bilirubin)

Thrombi use up platelets resulting in thrombocytopenia

Coagulation proteins are consumed resulting in elevated PT and PTT

*FDP and d-dimer are elevated as fibrinolysis occurs (d-dimer is the most sensitive but not specific test for DIC)



Released following plasmin-mediated degradation of fibrinogen/fibrin; d-dimer is specific fragment formed only upon degradation of cross-linked fibrin (after XIII works); more on this later

Lecture slide 36 - full slide retained. DIC creates fibrin-rich microthrombi that shear RBCs while simultaneously consuming platelets and coagulation proteins. PT/aPTT and D-dimer become major discriminators.

	DIC	TTP	Typical HUS
Microthrombus driver	Systemic coagulation activation / fibrin	Platelet-vWF microthrombi from severe ADAMTS13 deficiency	Endothelial injury, classically Shiga toxin
Platelets	Low	Low	Low
Schistocytes	Yes	Yes	Yes
PT/aPTT	Prolonged	Usually normal	Usually normal
Fibrinogen	Often low/consumed	Usually preserved	Usually preserved
Big clinical clue	Sepsis, trauma, malignancy/APL, obstetric catastrophe, toxin	Neuro findings; PLASMIC/high TTP suspicion	Child + bloody diarrhea + prominent renal failure
Treatment	Treat cause + supportive blood-component therapy as needed	Urgent plasma exchange ; do not wait for ADAMTS13	Supportive care in typical Shiga-toxin HUS

THROMBOTIC THROMBOCYTOPENIC PURPURA: PATHOLOGY

Remember ADAMTS13 cleaves very high weight vWf multimers

In TTP, patients usually develop an autoantibody against ADAMTS13

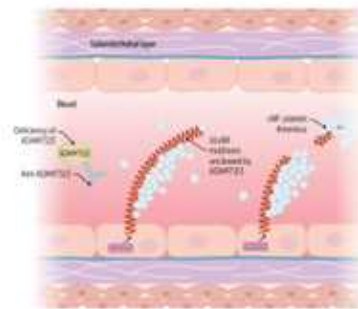
Very high weight multimers of vWf circulate and platelets adhere making microthrombi and resulting thrombocytopenia

Red cells get shredded on this – hemolysis with schistocytes/red cell fragments

Pentad: MAHA, thrombocytopenia, fever, renal abnormalities, and altered mental status

Some patients present with vague abdominal pain

Testing will show anemia and thrombocytopenia with schistocytes on smear, evidence of hemolysis (increased reticulocytes, elevated LDH, increased bilirubin, low haptoglobin), normal coagulation assays, and low ADAMTS13 activity (DO NOT WAIT FOR RESULTS OF THIS TEST)



Likelihood of TTP diagnosis can be calculated via PLASMIC score:

<https://www.mdcalc.com/plasmic-score-ttp>

Lecture slide 38 - full slide retained. TTP: autoantibody-mediated severe ADAMTS13 deficiency allows ultralarge vWf multimers to persist and collect platelets. PT/aPTT remain normal because coagulation factors are not being globally consumed.

THROMBOTIC THROMBOCYTOPENIC PURPURA: THERAPY

Plasma Exchange: initial therapy of choice

- Three goals:

- Removes antibody against ADAMTS13
- Removes circulating very high molecular weight vWf multimers
- Restores low levels of ADAMTS13 to normal

Generally fatal disease prior to use of plasma exchange (> 95% mortality)

- >80% survival with prompt treatment
- Successful outcome correlates with prompt diagnosis and initiation of treatment

Do not give platelets! They will continue to aggregate until underlying pathology addressed

Lecture slide 39 - full slide retained. Plasma exchange removes inhibitor and ultralarge multimers while restoring ADAMTS13. Savage emphasizes prompt treatment and explicitly warns against platelet transfusion before the underlying process is controlled.

EMERGENCY SPLIT

Schistocytes + low platelets + normal PT/aPTT -> TTP/HUS lane.

Schistocytes + low platelets + prolonged PT/aPTT + high D-dimer -> DIC lane.

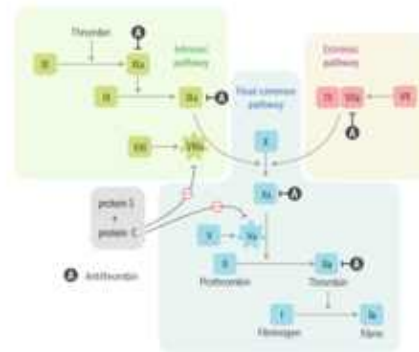
INHIBITION OF COAGULATION: KEY PLAYERS AND EVENTS

Clotting is great, but only when its not excessive or inappropriate; many things are involved in controlling it or inhibiting it

Antithrombin: inhibits all activated serine proteases (XI, X, IX, VII, II) with strongest effects on IIa and Xa; activity is increased with heparin

Protein C: inactivates V and VIII after thrombomodulin binds to thrombin (Batman)

Protein S: cofactor for C (Robin)



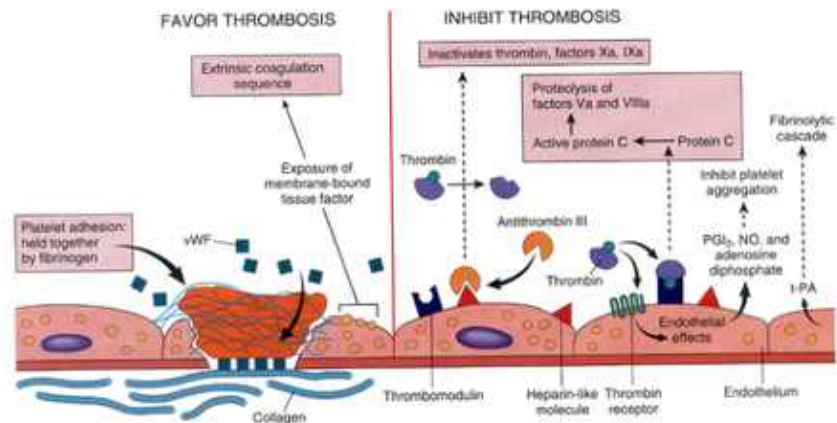
Lecture slide 42 - full slide retained. Antithrombin inhibits activated serine proteases with strong effects on IIa and Xa and is accelerated by heparin. Activated protein C inactivates Va and VIIIa; protein S is its cofactor.

THE BRAKE SYSTEM

Antithrombin -> IIa/Xa especially.

Protein C + Protein S -> turn off Va and VIIIa.

CONTROLLING HEMOSTASIS



Lecture slide 43 - full slide retained. This diagram places thrombomodulin/protein C, antithrombin, endothelial mediators, and fibrinolysis around the clot. It is the conceptual bridge from normal regulation to thrombophilia.

BOARD ENRICHMENT - WHY HEPARIN WORKS WITH ANTITHROMBIN

Heparin does not simply “thin blood.” It markedly accelerates **antithrombin** activity against clotting proteases. This is why antithrombin deficiency can cause relative heparin resistance.

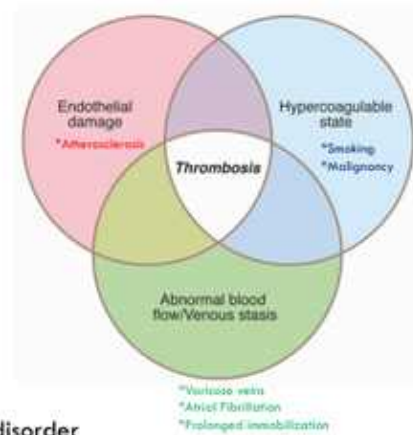
THROMBOPHILIA

Most disorders (not risk factors) are inherited

- Factor V Leiden (FVL)
- Prothrombin gene mutation
- Protein C deficiency
- Antithrombin deficiency – not covered today*
- Protein S deficiency
- *Antiphospholipid antibody syndrome (APS): acquired
- *Heparin induced thrombocytopenia (HIT): acquired

But most patients with thromboemboli do not have a disorder

Most patients do not require work-up



Lecture slide 45 - full slide retained. Savage emphasizes that most thromboemboli are explained by common acquired risk factors rather than a rare inherited disorder. The slide uses Virchow's triad: endothelial damage, hypercoagulability, and stasis.

FACULTY DIAGNOSTIC RESTRAINT

Savage explicitly says **most patients with thromboembolism do not require an inherited-thrombophilia work-up**. Testing can be expensive, may not change inpatient management, and some assays become unreliable during acute thrombosis or anticoagulant therapy.

Disorder	Mechanism	Key test / timing clue
Factor V Leiden	Factor V becomes resistant to activated protein C cleavage; slide specifies Arg506Gln.	Genetic testing; generally not an urgent inpatient test.
Prothrombin G20210A	3' UTR mutation -> increased prothrombin production.	Genetic test; result generally does not alter immediate inpatient therapy.
Protein C deficiency	Less inactivation of Va/VIIIa.	Do not test during acute clot/ warfarin if avoidable; levels are affected.
Protein S deficiency	Loss of protein C cofactor.	Same testing caveats; inherited or acquired.
Antithrombin deficiency	Reduced endogenous inhibition of thrombin/Xa and other proteases.	Listed in official objective; slide says not covered in depth.
APS / HIT	Acquired thrombophilic immune disorders.	Slide lists both; HIT expanded below from companion.

FACTOR V LEIDEN

Most common inherited thrombophilia -> factor V resists activated protein C.

Protein C/S and warfarin skin necrosis

Protein C and S are vitamin K-dependent and have short half-lives. Early warfarin therapy can reduce these natural anticoagulants before the procoagulant factors fully fall, creating a transient hypercoagulable period - especially dangerous in protein C/S deficiency.

Companion gap-fill

Heparin-induced thrombocytopenia (HIT)

LO 8 acquired thrombophilia

WHY THIS SECTION IS HERE

Savage lists HIT as an acquired thrombophilia on Slide 45 but does not develop it in the remaining deck. The uploaded companion gives the work-up and management details.

MECHANISM

IgG against heparin-PF4 complexes -> Fc receptor-mediated platelet activation -> thrombosis + platelet consumption. The key paradox is that the platelet count falls while the patient becomes **more thrombotic, not primarily more hemorrhagic.**

4 TS

Thrombocytopenia, Timing, Thrombosis, oTher causes.

Classic timing is around days 5-10 unless there has been recent prior heparin exposure.

Companion work-up / management

- High clinical probability -> **stop all heparin immediately, including flushes.**
- Start a **non-heparin anticoagulant** rather than waiting for confirmatory testing.
- PF4-heparin antibody immunoassay is a sensitive screen; a functional assay such as the **serotonin release assay** is confirmatory.
- Do not reflexively begin warfarin during acute HIT before platelet recovery.

HIT EXAM ANSWER

Low platelets after heparin + new thrombosis -> STOP heparin and anticoagulate with a non-heparin drug.

Bleeding patient

#	Question	Interpretation
1	Mucocutaneous/petechial or deep/delayed?	Mucocutaneous -> primary. Deep joints/muscles/delayed -> secondary.
2	Platelet count?	Low -> thrombocytopenia pathway. Normal -> qualitative platelet/vWF/factor problem.
3	PT and aPTT?	Localize extrinsic, intrinsic, common, or normal-assay disorders.
4	If prolonged, mixing study?	Corrects -> factor deficiency. Fails -> inhibitor.
5	Targeted test?	PFA/aggregation/vWF testing or factor assay/inhibitor titer.

Schistocytes + thrombocytopenia

PT/aPTT	Clinical clue	Think
Prolonged	Sepsis, trauma, APL, obstetric catastrophe; D-dimer high, fibrinogen low	DIC
Normal	Neuro-dominant, adult, ADAMTS13/PLASMIC	TTP
Normal	Child, bloody diarrhea, severe renal failure	Typical HUS

Thrombosis patient

DO NOT SHOTGUN THROMBOPHILIA PANELS

Start with **provoked vs unprovoked, age, recurrence, family history, unusual site, medications, malignancy, pregnancy/estrogen, immobilization, smoking.**

Inherited testing is selective and timing-sensitive.

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Draw adhesion-activation-aggregation with vWF/GpIb, ADP/P2Y12, GpIIb/IIIa/fibrinogen.
LO 2	Explain platelet count, MPV, PFA/bleeding time, and aggregation/ristocetin studies.
LO 3	Differentiate ITP, Glanzmann, Bernard-Soulier, and vWD using count/size/ristocetin/clinical bleeding.
LO 4	Know intrinsic/extrinsic/common pathways, II-VII-IX-X + C/S vitamin K dependence, and XIII's post-fibrin role.
LO 5	Interpret PT/aPTT and mixing studies, then choose factor assay vs inhibitor testing.
LO 6	Recognize Hemophilia A/B, isolated prolonged aPTT, correction with mixing, deep bleeding, and factor-specific assays.
LO 7	Separate DIC, TTP, and HUS using coagulation tests, clinical setting, renal/neuro pattern, and emergency treatment.
LO 8	Explain Factor V Leiden, prothrombin G20210A, protein C/S and antithrombin biology, plus selective work-up and acquired HIT logic.

LAST 60 SECONDS

Primary: vWF-GpIb adhesion -> activation -> GpIIb/IIIa-fibrinogen aggregation.

Secondary: PT = VII/common; aPTT = XII/XI/IX/VIII/common; mix corrects deficiency, fails inhibitor.

Hemophilia B: IX, X-linked, deep bleeding, PT normal, aPTT high, mixing corrects, IX low.

vWD: primary adhesion + VIII carrier.

DIC: schistocytes + low platelets + PT/aPTT high + D-dimer high.

TTP: ADAMTS13, schistocytes + low platelets + PT/aPTT normal -> plasma exchange now.

FVL: factor V resists protein C. **HIT:** low platelets but thrombosis -> stop heparin + non-heparin anticoagulant.

Source hierarchy: Savage's 50-slide Coagulation deck is the primary slide-order source. The official Week 7 objective sheet defines emphasis. The uploaded companion is supplemental, particularly for HIT and board-

style mechanisms. Current Hemophilia B therapy is clearly labeled external 2026 enrichment and linked to FDA sources. The personal dedication is intentionally brief and classroom-friendly.